

AI Agent Guide: Document AI

Piece: @activepieces/piece-document-ai
Action: document_ai
Version: ~0.0.1

What it does: Takes a document URL (PDF or image), extracts structured data using AI vision + processing models, and optionally maps results to a Data Board schema.

1. Input Settings

Property	Type	Required	Default	propertySettings type	Description
visionProvider	Dropdown	No	"system"	MANUAL	AI provider for document vision. "system" uses built-in models.
modelName	Dropdown	Yes	"gpt-4o"	MANUAL (resolve value via get_piece_options)	Vision model. Options depend on visionProvider.
additionalDocumentInstructions	LongText	No	""	MANUAL	Instructions for the vision/extraction phase. Use this to specify exact output field names.
url	ShortText	No	—	MANUAL	Direct URL to PDF or image. Required for single-document mode.
language	StaticDropdown	Yes	"English"	MANUAL	Output language. Options: "English", "Traditional Chinese", "Simplified Chinese".
insertToDataboard	Checkbox	No	true	MANUAL	If true, output keys follow board schema. Does NOT auto-insert data — you still need Automate Data Board piece.
boardId	Dropdown	No	—	MANUAL (resolve value via get_piece_options)	Target board. Only visible when insertToDataboard = true.
processingProvider	Dropdown	No	"system"	MANUAL	AI provider for processing/validation phase.
processModelName	Dropdown	Yes	—	MANUAL (resolve value via get_piece_options)	Processing model. Falls back to "gpt-4o" if "Default".
additionalInstructions	LongText	No	""	MANUAL	Instructions for the processing/validation phase.
utc	StaticDropdown	Yes	8	MANUAL	Timezone offset (UTC-12 to UTC+12).
timeout	Number	No	600	MANUAL	Per-request timeout in seconds.
maxConcurrent	Number	No	3	MANUAL	Max parallel requests for batch mode.
retryFailed	Checkbox	No	true	MANUAL	Retry failed requests with exponential backoff.
maxRetries	Number	No	2	MANUAL (with schema)	Max retry attempts. Only visible when retryFailed = true.

Conditional Visibility

- boardId is only visible when insertToDataboard == true
- maxRetries is only visible when retryFailed == true

2. ADD_ACTION — Add Document AI Step

Use this to add Document AI as a skeleton step. You MUST set name to an explicit step name.

```

Update_flow(
  flow_id = "<flow_id>",
  operation = "ADD_ACTION",
  operation_body = {
    "parentStep": "trigger",
    "stepLocationRelativeToParent": "AFTER",
    "branchIndex": 0,
    "action": {
      "name": "step_1",
      "displayName": "Extract from Document",
      "type": "PIECE",
      "valid": false,
      "settings": {
        "pieceName": "@activepieces/piece-document-ai",
        "pieceVersion": "~0.0.1",
        "actionName": "document_ai",
        "input": {},
        "propertySettings": {},
        "errorHandlingOptions": {}
      }
    }
  }
)

```

3. UPDATE_ACTION — Configure Document AI (insertToDataboard = true) — RECOMMENDED

❑ **ALWAYS use this config when a Data Board step exists in the flow.** Setting `insertToDataboard=true + boardId` ensures the output field names match the board's field schema, which is required for downstream field mappings to work.

Output fields are ALWAYS at `data.xxx` directly (NO `properties` wrapper). When `insertToDataboard=true`, field names match the board schema. When `false`, field names match `additionalDocumentInstructions`.

Before calling UPDATE_ACTION:

1. Call `get_piece_options` to resolve `modelName` (`refresher: visionProvider`)
2. Call `get_piece_options` to resolve `processModelName` (`refresher: processingProvider`)

```

Update_flow(
  flow_id = "<flow_id>",
  operation = "UPDATE_ACTION",
  operation_body = {
    "name": "step_1",
    "displayName": "Extract from Business Card",
    "type": "PIECE",
    "valid": true,
    "skip": false,
    "settings": {
      "pieceName": "@activepieces/piece-document-ai",
      "pieceVersion": "~0.0.1",
      "actionName": "document_ai",
      "input": {
        "visionProvider": "system",
        "modelName": "gpt-4o",
        "additionalDocumentInstructions": "You are extracting business card information. Return a JSON object with keys exactly: full",
        "url": "{{trigger.body['file_url']}}",
        "language": "English",
        "insertToDataboard": true,
        "boardId": "<resolved_board_id_from_get_piece_options>",
        "processingProvider": "system",
        "processModelName": "gpt-4o",
        "additionalInstructions": "Validate and normalize fields. Ensure email and website are well-formed if present. Do not halluc",
        "utc": 8,
        "timeout": 600,
        "maxConcurrent": 3,
        "retryFailed": true,
        "maxRetries": { "maxRetries": 3 }
      },
    },
    "propertySettings": {
      "visionProvider": { "type": "MANUAL" },
      "modelName": { "type": "MANUAL" },
      "additionalDocumentInstructions": { "type": "MANUAL" },
      "url": { "type": "MANUAL" },
      "language": { "type": "MANUAL" },
      "insertToDataboard": { "type": "MANUAL" },
      "boardId": { "type": "MANUAL" },
      "processingProvider": { "type": "MANUAL" },
      "processModelName": { "type": "MANUAL" },
      "additionalInstructions": { "type": "MANUAL" },
      "utc": { "type": "MANUAL" },
      "timeout": { "type": "MANUAL" },
      "maxConcurrent": { "type": "MANUAL" },
      "retryFailed": { "type": "MANUAL" },
      "maxRetries": {
        "type": "MANUAL",
        "schema": {
          "maxRetries": {
            "displayName": "Max Retries",
            "required": false,
            "defaultValue": 2,
            "type": "NUMBER"
          }
        }
      }
    },
  },
  "errorHandlingOptions": {
    "continueOnFailure": { "value": false },
    "retryOnFailure": { "value": false }
  }
}
)

```

4. UPDATE_ACTION — Configure Document AI (insertToDataboard = false) — NOT RECOMMENDED

⚠ **Only use this when NO Data Board step exists in the flow.** Output field names come from `additionalDocumentInstructions` instead of the board schema.

If a Data Board step exists, use Section 3 instead (`insertToDataboard = true`).

```

Update_flow(
  flow_id = "<flow_id>",
  operation = "UPDATE_ACTION",
  operation_body = {
    "name": "step_1",
    "displayName": "Extract from Document",
    "type": "PIECE",
    "valid": true,
    "skip": false,
    "settings": {
      "pieceName": "@activepieces/piece-document-ai",
      "pieceVersion": "~0.0.1",
      "actionName": "document_ai",
      "input": {
        "visionProvider": "system",
        "modelName": "gpt-4o",
        "additionalDocumentInstructions": "Extract all relevant fields from this document.",
        "url": "{{trigger.body['file_url']}}",
        "language": "English",
        "insertToDataboard": true,
        "boardId": "<resolved_board_id_from_get_piece_options>",
        "processingProvider": "system",
        "processModelName": "gpt-4o",
        "additionalInstructions": "Validate extracted fields against board schema.",
        "utc": 8,
        "timeout": 600,
        "maxConcurrent": 3,
        "retryFailed": true,
        "maxRetries": { "maxRetries": 3 }
      },
      "propertySettings": {
        "visionProvider": { "type": "MANUAL" },
        "modelName": { "type": "MANUAL" },
        "additionalDocumentInstructions": { "type": "MANUAL" },
        "url": { "type": "MANUAL" },
        "language": { "type": "MANUAL" },
        "insertToDataboard": { "type": "MANUAL" },
        "boardId": { "type": "MANUAL" },
        "processingProvider": { "type": "MANUAL" },
        "processModelName": { "type": "MANUAL" },
        "additionalInstructions": { "type": "MANUAL" },
        "utc": { "type": "MANUAL" },
        "timeout": { "type": "MANUAL" },
        "maxConcurrent": { "type": "MANUAL" },
        "retryFailed": { "type": "MANUAL" },
        "maxRetries": {
          "type": "MANUAL",
          "schema": {
            "maxRetries": {
              "displayName": "Max Retries",
              "required": false,
              "defaultValue": 2,
              "type": "NUMBER"
            }
          }
        }
      },
      "errorHandlingOptions": {
        "continueOnFailure": { "value": false },
        "retryOnFailure": { "value": false }
      }
    }
  }
)

```

5. Output Schema

The action returns an object with two arrays: `success` and `error`.

5.1 Full Output Structure

```
{
  "success": [
    {
      "success": true,
      "data": {
        "full_name": "Chucky Norris",
        "company": "Greenhouse Design",
        "job_title": "Interior Designer",
        "email": "chucky@greenhouse.com.au",
        "phone": {
          "country_code": "AU",
          "country_calling_code": "61",
          "phone": "246178827",
          "calling_code_with_number": "+61246178827",
          "national_number": "246178827"
        },
        "address": "123 Main St, Sydney",
        "website": "http://www.greenhouse.com.au",
        "raw_ocr_text": "Full OCR text of the document..."
      }
    }
  ],
  "error": []
}
```

5.2 Correct Access Path

ALL extracted fields are DIRECTLY inside `data`, NOT inside `data.properties`.

WRONG: `{{step_1['success'][0]['data']['full_name']}}` ← `properties` does NOT exist
CORRECT: `{{step_1['success'][0]['data']['full_name']}}`

❑ There is NO `properties` wrapper in Document AI output. Fields are at `data.xxx` directly.

5.3 Access Patterns by Field Type

Field Type	Access Path	Returns
String (name, company, email, etc.)	<code>{{step_1['success'][0]['data']['full_name']}}</code>	"Chucky Norris"
Phone (object)	<code>{{step_1['success'][0]['data']['phone']}}</code>	<code>{"country_code": "AU", "phone": "246178827", ...}</code>
Phone (string value)	<code>{{step_1['success'][0]['data']['phone']['calling_code_with_number']}}</code>	"+61246178827"
Phone (national)	<code>{{step_1['success'][0]['data']['phone']['national_number']}}</code>	"246178827"
Raw OCR text	<code>{{step_1['success'][0]['data']['raw_ocr_text']}}</code>	Full text string

5.4 Output Behavior by Configuration

Configuration	Output behavior
<code>insertToDataboard = false</code>	<code>data</code> keys match the field names you specified in <code>additionalDocumentInstructions</code> .
<code>insertToDataboard = true + boardId</code>	set <code>data</code> keys match the board's field names . Data is also inserted into the board.
❑ CRITICAL: When <code>insertToDataboard=true</code> , output field names come from the BOARD , not from <code>additionalDocumentInstructions</code> . If the board has a field named <code>name</code> (not <code>full_name</code>), the output will be <code>data['name']</code> , NOT <code>data['full_name']</code> . Always check the board's actual field names and use those in downstream references (Return Response, Gmail, etc.).	

Example: If board fields are `Full Name`, `Email`, `Phone`, `Company`:

```
WRONG:  {{step_1['success'][0]['data']['full_name']}}    ← doesn't match board field name
CORRECT: {{step_1['success'][0]['data']['Full Name']}}    ← matches board field exactly
```

6. Wiring to Automate Data Board

Fields are directly in data, NOT in data.properties. Map the **Identifier** field to a trigger value (NOT an OCR field) to avoid empty identifier errors.

```

Update_flow(
  flow_id = "<flow_id>",
  operation = "UPDATE_ACTION",
  operation_body = {
    "name": "step_2",
    "displayName": "Create Databoard Record",
    "type": "PIECE",
    "valid": true,
    "skip": false,
    "settings": {
      "pieceName": "@activepieces/piece-automate-data-board",
      "pieceVersion": "~0.0.1",
      "actionName": "auto_data_board",
      "input": {
        "workflowName": "Automate Data Board",
        "version": "1",
        "boardId": "<board_id>",
        "chooseAction": "create",
        "inputMethod": { "inputMethod": "json" },
        "contentJson": {
          "contentJson": {
            "<identifier_field_id>": "{{trigger.body['file_url']}}",
            "<full_name_field_id>": "{{step_1['success'][0]['data']['full_name']}}",
            "<company_field_id>": "{{step_1['success'][0]['data']['company']}}",
            "<email_field_id>": "{{step_1['success'][0]['data']['email']}}",
            "<phone_field_id>": "{{step_1['success'][0]['data']['phone']['calling_code_with_number']}}",
            "<job_title_field_id>": "{{step_1['success'][0]['data']['job_title']}}",
            "<website_field_id>": "{{step_1['success'][0]['data']['website']}}",
            "<address_field_id>": "{{step_1['success'][0]['data']['address']}}",
            "<raw_ocr_field_id>": "{{step_1['success'][0]['data']['raw_ocr_text']}}",
            "<sent_to_email_field_id>": "{{trigger.body['target_email']}}"
          }
        },
        "testing_organization_id": "<org_id>"
      },
      "propertySettings": {
        "workflowName": { "type": "MANUAL" },
        "version": { "type": "MANUAL" },
        "boardId": { "type": "MANUAL" },
        "chooseAction": { "type": "MANUAL" },
        "inputMethod": {
          "type": "MANUAL",
          "schema": {
            "inputMethod": {
              "displayName": "Input Method",
              "required": true,
              "options": {
                "options": [
                  { "label": "Fields", "value": "field" },
                  { "label": "JSON", "value": "json" }
                ]
              },
              "type": "STATIC_DROPDOWN"
            }
          }
        },
        "contentJson": {
          "type": "MANUAL",
          "schema": {
            "contentJson": {
              "displayName": "Content (JSON)",
              "required": true,
              "type": "JSON"
            }
          }
        }
      }
    }
  }
)

```



```

    },
    "testing_organization_id": {
      "type": "MANUAL",
      "schema": {
        "testing_organization_id": {
          "displayName": "Organization ID",
          "required": false,
          "type": "SHORT_TEXT"
        }
      }
    }
  },
  "errorHandlingOptions": {
    "continueOnFailure": { "value": false },
    "retryOnFailure": { "value": false }
  }
}
)

```

Key Rules for Data Board Wiring:

- Use **field IDs** (24-char hex like 69cdf142...) as keys, NOT field names.
- The **identifier field** (`is_identifier: true`) **MUST** have a non-empty value. Use `{{trigger.body['file_url']}}` or another trigger value — NEVER use an OCR field as the sole identifier.
- Phone fields return an object. Use `['calling_code_with_number']` or `['national_number']` to get a string.
- All board field types (ShortText, LongText, Email, Phone, Link, etc.) expect **string values**.

7. Wiring to Gmail

```

Update_flow(
  flow_id = "<flow_id>",
  operation = "UPDATE_ACTION",
  operation_body = {
    "name": "step_3",
    "displayName": "Send OCR Results Email",
    "type": "PIECE",
    "valid": true,
    "skip": false,
    "settings": {
      "pieceName": "@activepieces/piece-gmail",
      "pieceVersion": "~0.9.6",
      "actionName": "send_email",
      "input": {
        "receiver": [{"trigger.body['target_email']}"],
        "cc": [],
        "bcc": [],
        "subject": "Your business card OCR results",
        "body_type": "plain_text",
        "body": "Here are your business card OCR results:\n\nFull Name: {{step_1['success'][0]['data']['full_name']}}\nCompany: {{step_1['success'][0]['data']['company']}}",
        "reply_to": [],
        "sender_name": "Business Card OCR Bot",
        "from": "me",
        "attachments": [],
        "in_reply_to": "",
        "draft": false
      },
      "propertySettings": {
        "receiver": { "type": "MANUAL" },
        "cc": { "type": "MANUAL" },
        "bcc": { "type": "MANUAL" },
        "subject": { "type": "MANUAL" },
        "body_type": { "type": "MANUAL" },
        "body": { "type": "MANUAL" },
        "reply_to": { "type": "MANUAL" },
        "sender_name": { "type": "MANUAL" },
        "from": { "type": "MANUAL" },
        "attachments": { "type": "MANUAL" },
        "in_reply_to": { "type": "MANUAL" },
        "draft": { "type": "MANUAL" }
      },
      "errorHandlingOptions": {
        "continueOnFailure": { "value": false },
        "retryOnFailure": { "value": false }
      }
    },
    "connections": {
      "@activepieces/piece-gmail": "<gmail_connection_id>"
    }
  }
)

```

Note: Gmail requires an active OAuth connection. Resolve via `get_connections`. Set `"from": "me"` to use the authenticated account's email.

8. Wiring to Return Response

```

Update_flow(
  flow_id = "<flow_id>",
  operation = "UPDATE_ACTION",
  operation_body = {
    "name": "step_4",
    "displayName": "Return Response",
    "type": "PIECE",
    "valid": true,
    "skip": false,
    "settings": {
      "pieceName": "@activepieces/piece-webhook",
      "pieceVersion": "~0.1.24",
      "actionName": "return_response",
      "input": {
        "responseType": "json",
        "fields": {
          "status": 200,
          "headers": {},
          "body": {
            "success": true,
            "full_name": "{{step_1['success'] [0] ['data'] ['full_name']}}",
            "company": "{{step_1['success'] [0] ['data'] ['company']}}",
            "email": "{{step_1['success'] [0] ['data'] ['email']}}",
            "phone": "{{step_1['success'] [0] ['data'] ['phone'] ['calling_code_with_number']}}",
            "job_title": "{{step_1['success'] [0] ['data'] ['job_title']}}",
            "website": "{{step_1['success'] [0] ['data'] ['website']}}",
            "address": "{{step_1['success'] [0] ['data'] ['address']}}",
            "raw_ocr_text": "{{step_1['success'] [0] ['data'] ['raw_ocr_text']}}",
            "record_id": "{{step_2['success'] ['recordId']}}"
          }
        },
        "respond": "stop"
      },
      "propertySettings": {
        "responseType": { "type": "MANUAL" },
        "fields": {
          "type": "MANUAL",
          "schema": {
            "status": { "displayName": "Status", "required": false, "defaultValue": 200, "type": "NUMBER" },
            "headers": { "displayName": "Headers", "required": false, "type": "OBJECT" },
            "body": { "displayName": "JSON Body", "required": true, "type": "JSON" }
          }
        },
        "respond": { "type": "MANUAL" }
      },
      "errorHandlingOptions": {
        "continueOnFailure": { "value": false },
        "retryOnFailure": { "value": false }
      }
    }
  }
)

```

9. Rules & Gotchas

9.1 Identifier Field Must Not Be Empty

Databoard boards have an identifier field (`is_identifier: true`). If this field's value is "", the API returns 400: "Identifier value is required". ALWAYS map the identifier field to a value guaranteed to be non-empty:

- **SAFE:** `{{trigger.body['file_url']}}` (always present from webhook)
- **UNSAFE:** `{{step_1['success'] [0] ['data'] ['full_name']}}` (OCR may return "")

9.2 All Board Fields Expect Strings

ShortText, LongText, Email, Phone, Link — all expect string values.

- **WRONG:** "<field_id>": "{{step_1['success'] [0] ['data'] }}" → **passes** [object Object]
- **CORRECT:** "<field_id>": "{{step_1['success'] [0] ['data'] ['full_name'] }}" → **passes** "Chucky Norris"

9.3 Phone Field Returns Object

Document AI returns phone as an object with sub-fields. For board Phone fields, use a string sub-key:

- {{step_1['success'] [0] ['data'] ['phone'] ['calling_code_with_number'] }} → "+61246178827"
- {{step_1['success'] [0] ['data'] ['phone'] ['national_number'] }} → "246178827"

9.4 No Handlebars Conditionals

ActivePieces piece inputs do NOT support {{#if}} / {{else}} blocks. They concatenate both branches.

- **WRONG:** "{{#if step_1['error'].length}}failed{{else}}success{{/if}}" → **produces** "failedsuccess"
- **CORRECT:** Use a Code step for conditional logic, then reference the Code step's output.

9.5 insertToDataboard Does NOT Insert

Setting `insertToDataboard = true` only makes Document AI use the board's field schema for extraction output. You **MUST** still add a separate Automate Data Board step to actually create/update records.

9.6 Recommended Flow Pattern

1. Webhook trigger → receives `file_url` + `target_email`
2. Document AI (step_1) → `insertToDataboard=true` + `boardId` → output at `step_1['success'] [0] ['data']`
3. Automate Data Board (step_2) → creates record using field IDs + data path
4. Gmail (step_3) → sends email with extracted data
5. Return Response (step_4) → returns JSON with `record_id` + extracted fields

❑ **Document AI MUST have `insertToDataboard=true` + `boardId` set when Data Board step exists.** This ensures output field names match the board schema for consistent downstream mappings.

❑ **Output is ALWAYS at `data.xxx` directly — there is NO `properties` wrapper.**

Map identifier field to {{trigger.body['file_url']}} (always non-empty). Map all OCR fields via {{step_1['success'] [0] ['data'] ['<board_field_name>'] }}.

Use the EXACT board field names (case-sensitive, may contain spaces like Full Name).